

From Gestural Handshapes MAN & WOMAN to Agreement: Derivational Pathways in Japanese Sign Language

Abstract:

This paper revisits the gendered handshape system in Japanese Sign Language (JSL), where the “TA” (thumb, also referred as “タ” or MAN) and “I” (pinky finger, also referred to as “イ” or WOMAN) originated as culturally rooted gestures (see the note 1 for the gestural handshapes) during Edo period (1603-1868). Building on Osugi & Supalla’s (1988) structural analysis, we aim to organize and review the pronominal system of JSL, trace its historical development. This analysis draws on existing elicited JSL data (Tamura, 2025; Osugi & Supalla, 1988; Tomita, 2021).

Notably, while prior research has highlighted verb agreement and classifier predicates as central to signed languages, often emphasizing noun-to-verb derivation tendencies in signed languages, JSL presents a distinctive trajectory: It begins with development of a pronominal system from gestural origins, followed by the expansion into nominal expressions, and then progresses toward the formation of verb structures and agreement construction. These handshapes continually appeared and evolved into grammatical markers through systematic use, illustrating the modality-specific pathways of morphosyntactic development.

We propose a five-stage derivational sequence in JSL:

- 1) **Nominalization from lexical roots**, using TA and I handshapes marking gender distinctions.
- 2) **Development of a pronominal system**, with these handshapes in combination with spatial loci or movements to indicate grammatical roles (Figure 1: JSL pronominal system).
- 3) **Expansion into semantically complex words**, leveraging productive combinations of phonological features and handshapes (Figure 2: GREETING, MEETING, MEN).
- 4) **Derivation from nouns to verbs**, using movement and spatial modification (Figure 3: Derivation from nouns to verbs: GET-MARRIED, DIVORCE, and APPROACH).
- 5) **Integration into agreement verbs**, where the handshapes go directionality and argument structure within verb constructions (Figure 4: CALL with the hand of I “イ” handshape).

The five-stage process reveals how modality-specific factors interact with usage-based frequency effects (Bybee 2006), fostering the development of handshapes as functional morphemes. Our data illustrate how handshape and movement combinations stabilize over time into grammatical constructions, maintaining the productivity while simultaneously functioning as frozen forms. This study contributes to cross-modality perspectives on grammaticalization (Sandler 2016), demonstrating that while iconicity and cultural knowledge initially guide handshape selection, the repeated use within specific syntactic and discourse contexts increases arbitrariness and grammatical stability. Additionally, the JSL case exemplifies how cultural factors (e.g., gendered handshapes rooted in Japanese cultural gestures) and cognitive constraints shape the morphosyntactic development of sign languages.

In comparing JSL with American Sign Language (ASL) and Israeli Sign Language (ISL), we argue that JSL’s five-stage derivational sequence challenges assumptions about signed language morphology. Prior work often considers verb directionality and agreement as central to morphological structures, while also regarding classifier predicates as a core of morphological and syntactic structure. The JSL pathway suggests the need to reassess these perspectives, highlighting the importance of the national gesture system, and frequent usage fosters morphosyntactic innovation within language. It is particularly intriguing that, from the perspective of productivity, the TA(タ) and I(イ) handshapes in JSL maintain a level of productivity while simultaneously functioning as frozen forms, illustrating the dynamic interplay between fixedness and creativity in the language.

Figure 1: JSL pronominal system

Classification	Handshape	Meaning	Function & Details
No Gender Marker		YOU I	Indicates deictically; no gender information
Referential function		HE SHE	Forefinger +referential Uses Locus to specify meaning by context. Depending on verbs, the handshape TA and I is used continuously within the signing space. Verb type may change the case of the pronoun; possible use as relative pronoun
Anaphoric function with movement		X's male partner X's female partner	Functions as a possessive pronoun
Plural emphasis		THEY WE	Plural and deictic and referential
Case Expression		[to X] [from X]	The closer location to the signer functions as the subjects, farther location as the object
Use in Kinship Contexts		[X's father] etc	Functions as a personal pronoun; grammatical case is determined by location. People of higher status are placed higher/gender information may be indicated/ pointing required before referencing kinship terms
Adjustment for Politeness and Respect		[Polite suffix]	Adjustment are made through spatical position and handshapes

Note 1: gendered handshape MAN and WOMAN are represented in this picture



Figure 2: Expansion into semantically complex words (GREETING, MEETING, MEN)

JSL GLOSS	Picture of the sign	Phonological description
GREETING		The thumbs of the TA handshape move downward, mimicking the gesture of a person lowering their head in greeting.
MEETING		The TA handshapes of both hands collide simultaneously. The contact point is specifically the area between the MTP joints.
MEN		The TA handshapes of both hands trace a circular path while approaching each other near the signer, indicating gathering or coming together.

Note 2: “MTP” (metacarpophalangeal) is used for anatomical precision when describing the contact point.

Figure 3: Derivation from nouns to verbs: GET-MARRIED, DIVORCE, and APPROACH

JSL GLOSS	Picture of the sign	Phonological description
GET-MARRIED		The TA handshape is placed at a distant location and then moves to connect.
DIVORCE		The TA handshape, which was connected, separates while the mouth simultaneously opens. Often used when talking about past events, so mouth opening may not always occur.
1. APPROACH from a woman to man 2. APPROACH from a man to woman		This expresses relationship dynamics. 1. The TA handshape remains in a signing space, while I handshape is placed at distance. The hand with I handshape then moves toward the TA hand at the location. 2. The hand representing the TA handshape moves toward the other hand which representing I handshape.

Figure 4: Agreement verb CALL with the hand of I (I) handshape



References

Osugi, Y., & Supalla, T. (1998). [A structural analysis of gendered handshapes in Japanese Sign Language]. *Sign Language Communication Studies*, 27, 60–74. (In Japanese).